



High-speed electric trains

Birmingham Airport Maglev UK 1984



26 mph (42 km/h)

Train is held $\frac{1}{2}$ in (15 mm) above the track by magnets

VT Class Pendolino UK 2002



140 mph (225 km/h)

Cabin tilts up to 8 degrees when traveling on bends

DB ICE 3 Germany 2000



199 mph (320 km/h)

Engineer's cab separated from passenger seats by a glass panel

Shanghai Transrapid Maglev China 2004



249 mph (400 km/h)

Magnets raise the train around $\frac{3}{4}$ in (10 mm) above the guideway

199 mph (320 km/h)

SNCF TGV POS France 2006



The need for speed has never been greater as high-speed trains take on aircraft and road traffic to get passengers from one point to another in the quickest possible time. Meet some of the most rapid railroad vehicles of all time.

The superfast **JRN700 Shinkansen** train can accelerate from 0 to 168 mph (270 km/h) in three minutes and can tilt slightly to keep its speed when moving around bends. While most high-speed electric trains, such as the **Hyundai Rotem KTX**, have powerful wheel-turning